FACULTY OF COMPUTING

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SECP3843 – SPECIAL TOPIC IN DATA ENGINEERING

SECTION 1

TECHNICAL REPORT

TITLE:

OLTP (ONLINE TRANSACTION PROCESSING) VS OLAP (ONLINE ANALYTICAL PROCESSING)

GROUP 9

STUDENT NAME	MATRIC NUMBER
CHUA JIA LIN	A23CS0069
CHERYL CHEONG KAH VOON	A23CS0060

LECTURER: DR. ARYATI BINTI BAKRI

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EXECUTIVE SUMMARY

The article discusses the distinctions between advanced data management tools which is Online Transaction Processing (OLTP) and Online Analytical Processing (OLAP). The OLTP system is responsible for real-time transactions management, whereas the OLAP system is responsible for decision making using past data.

Besides, the analysis highlights that OLTP and OLAP perform best when working together although they are good at different aspects. OLTP ensures the stability of business operations, OLAP ensures meaningful insights to be generated. Furthermore, this research evaluates the advantages, challenges, and real-world applications of OLTP and OLAP technologies.

According to the report, OLTP and OLAP systems should be combined through data warehouses and cloud technologies in order to improve the operational efficiency and achieve long-term organization growth.

INTRODUCTION

In today's digital economy, data processing system play an important role in supporting daily operations and strategic decision-making by managing the large volume of data through transactions such as sales, payments and customer interaction. To manage this data efficiently, different types of processing systems are used which are Online Transaction Processing (OLTP) and Online Analytical Processing (OLAP).

According to Elmasri and Navathe (2016), OLTP systems are designed to handle real-time transactional data, focusing on the "current" state of business operations by prioritizing high concurrency and fast response times for simple, repetitive queries. These systems rely on ACID properties (Atomicity, Consistency, Isolation, and Durability) to guarantee that every business transaction is processed accurately and securely, even in the event of system failure.

On the other hand, OLAP systems are used to analyze historical data and support high-level decision making by processing complex multidimensional queries that aggregate huge amounts of information. These analytical queries are typically executed in a separate data warehouse

environment that is isolated from the real-time OLTP database since they are computationally intensive and can degrade system performance.

This report aims to compare OLTP and OLAP in terms of their functions, characteristics, advantages and application. Furthermore, it highlights how both systems complement each other through integration strategies to provide a competitive advantage in modern data- driven environments.

BACKGROUND / LITERATURE REVIEW

Online Transaction Processing (OLTP) systems are widely recognized as the foundation of operational information systems. According to Silberschatz et al. (2001), OLTP systems are designed to support real-time transaction processing, handling large volume of short and simple queries efficiently. These systems are typically implemented using relational databases and are optimized for speed, concurrency and data integrity.

According to Elmasri and Navathe (2015), importance of OLTP systems in industries such as banking, healthcare and e-commerce, where accurate and immediate data processing is essential. OLTP systems manage transactional activities such as payments, booking and inventory updates ensuring that data is consistently updated in real-time. To maintain reliability, OLTP systems enforce strict ACID properties which guarantee data consistency and accuracy (Silberschatz et al., 2001).

However, OLTP systems are not suitable for complex analytical queries or long-term data analysis. As a result, recent studies emphasize the need to integrate OLTP with analytical systems to support both operational efficiency and strategic decision-making (Amazon Web Services, 2023).

The Online Analytical Processing (OLAP) uses multidimensional data models or simply known as data cubes. Abdulghani (2008) states that data cubes are the basic OLAP structures. It arranges and analyzes data across different dimensions such as time and location. These data cubes pre-analyze and store the aggregated data so that the query performance could significantly enhanced and ease for complex analysis in future.

In addition, the analytical operations such as slicing, drill-down, and roll-up allows users to explore data interactively. Chaudhuri and Dayal (1997) claim that these operations are very flexible and

enable analysts to analyze data at different hierarchy levels for detailed analysis. This capability is very useful in business intelligence field, because decision makers can easily identify popular trends.

ANALYSIS & DISCUSSION

Online Transaction Processing (OLTP)

OLTP system process transactions in real-time, ensuring immediate data changes and consistent records. According to Connolly & Begg (2015), they use relational databases designed specifically for fast queries and concurrent access. Transactions such as INSERT, UPDATE and DELETE are recorded instantly through indexes and caching, reducing response time and allowing multiple users to operate simultaneously without affecting data integrity. (Ramakrishnan & Gehrke, 2003).

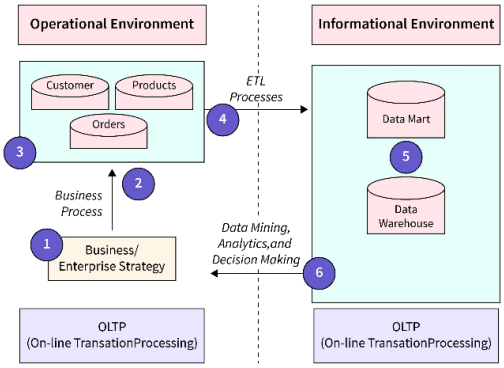

Figure 1 OLTP Process

OLTP is widely used in the banking industries such as ATMs, online banking and payment systems, where speed and accuracy are critical (Elmasri & Navathe, 2015). E-commerce platforms like Shopee rely on OLTP to update orders, inventory and payments instantly. Retail POS systems and hospital management systems also use OLTP for registration and appointment handling (Silberschatz et al., 2001).

OLTP system designed to handle real-time transaction quickly accurately, making them suitable for daily business operations and environments with many users. They use relational database that store atomic data are optimized for fast query performance. However, their performance may decline as data volume increase and managing high transaction loads while maintaining ACID compliance and system are not suitable for complex analytical queries. Due to these limitation, OLTP system are often separated from OLAP systems, which are designed for data analysis (Conn,

2005). To address these challenges, emerging trends such as hybrid transactional/analytical processing (HTAP) and cloud-based OLTP solution are being developed to improved scalability and support real-time analytics (Nayak, 2025).

Online Analytical Processing (OLAP)

Figure 2 demonstrates the process of OLAP. The process starts by extracting data from multiple data sources and store the extracted data are stored in a centralized storage known as data warehouse, which supports analytical processes. After that, the data is then organized into multidimensional cubes that eases the aggregation and analysis processes across multiple dimensions to improve query performance. Lastly, users perform operations such as slicing, dicing, drill-down, and roll-up using the data cubes to analyse the data when needed.

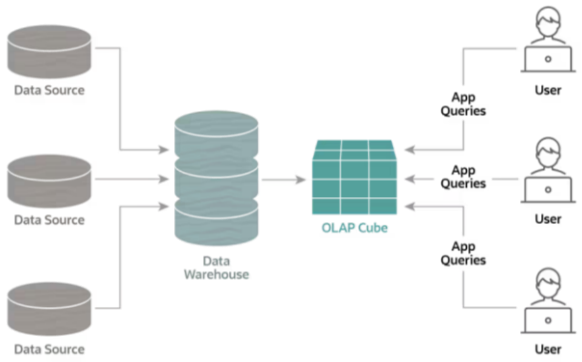

Figure 2 OLAP Process (NetSuite.com, 2022)

After OLAP was introduced, OLAP systems are widely used in inventory management to improve decision making and operational efficiency. It allows organizations to analyze inventory data across multiple dimensions such as time, product, and location (Exploring the Top OLAP Use Cases for Effective Data Analysis, 2024). As a result, businesses can monitor their inventory levels and identify trends easily to prevent overstocking or understocking problems.

The benefits of OLAP systems are quite significant. It has the ability to quickly process those complicated queries and facilitate data analysis from large amounts of data. Additionally, OLAP also enhances decision making by offering insightful information from past data. However, apart from these benefits, OLAP systems also face some challenges. Since OLAP uses data warehouses,

so it requires a large amount of storage. Besides, it also requires more processing power to handle large and multidimensional data (Abdulghani, 2008).

RECOMMENDATIONS

The company needs a data architecture model that separates OLTP and OLAP systems. The systems will be guaranteeing efficiency and performance. Businesses use online transaction processing systems to manage real-time transaction processing and operations. The company are able to keep accurate and consistent transaction records by maintaining positive ACID compliance. For example, the banking, finance and even online shopping websites use it because of their critical nature. In other words, they cannot afford the luxury of unstable systems. The company should establish an organized data warehouse to build the system architecture. Company can utilize ETL processes to extract data from OLTP source and load it into OLAP data models. These organizations can easily evaluate the data without impacting the business activities. Moreover, cloud platforms such as AWS or Azure can be used to improve scalability, availability and overall system flexibility to handle growing data volume. This approach enables organizations to support long-term strategic decision-making while maintaining efficient operations.

For OLAP systems, there are some recommendations. First, companies should build strong data warehouses that can support complex OLAP operations. According to Inmon (2005), a well-designed data warehouse organizes the data consistently and prepare it for analysis. This improves decision-making skills because the organization can get precise insights more easily. Additionally, organizations should integrate OLAP with OLTP systems for better performance. According to Chaudhuri and Dayal (1997), the OLTP systems are good at handling daily operations like transactions, while OLAP systems are better for analyzing data and strategic planning. When companies connect both systems, they can manage their organization efficiently and get valuable insights for long-term planning.

CONCLUSION

In conclusion, OLTP and OLAP systems play an important role in the management and utilization of data in an organization. The OLTP systems help organizations run their daily business operations smooth, while OLAP systems help them to understand trends and make better business decisions.

Although each system has its own advantages and limitations, but integrating both systems together is a great idea because organizations can maintain high performance and obtain meaningful insights in this way by separating operational and analytical processes.

Overall, the integration of OLTP and OLAP systems is really important for organizations that want to stay competitive in this data-driven world. With strong data warehouses and cloud computing, companies can improve their efficiency and decision-making for long-term success.

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LOGBOOK

Date	Student	Activity	Prompt Used	References	Verification
1/4/2026	Cheryl, Chua Jia Lin	Divide tasks and discuss selected topic with groupmate.	N/A	N/A	N/A
3/4/2026	Chua Jia Lin	Search for OLAP information using ChatGPT, Google Scholar, and Scopus AI for Literature Review part.	"OLAP", "OLAP Literature Review"	1. Abdulghani, A. A. (2009). Computation of OLAP data cubes. In *IGI Global eBooks* (pp. 286–292). https://doi.org/10.4018/978-1-60566-010-3.ch046 2. Chaudhuri, S., & Dayal, U. (1997). An overview of data warehousing and OLAP technology. *ACM SIGMOD Record*, *26*(1), 65–74. https://doi.org/10.1145/248603.248616	Complete OLAP Literature Review part based on key points obtained from AI tools.
4/42026	Cheryl Cheong Kah Voon	Search for OLTP information using ChatGPT, Gemini, Google Scholar for Introduction part and Background/Literature Review	"What is OLTP and OLAP"	1. Elmasri, R., & Navathe, S. B. (2016). Fundamentals of database systems seventh edition. 2. Conn, S. S. (2005, April). OLTP and OLAP data integration: a review of feasible implementation methods and architectures for real time data analysis. In *Proceedings. IEEE SoutheastCon, 2005.* (pp. 515-520). IEEE.	Cross-checked definitions of ACID properties and the separation of OLTP/OLAP environments against the textbook *Fundamentals of Database Systems* (7th ed.) to ensure technical accuracy

4/4/2026	Chua Jia Lin	Search for OLAP information using ChatGPT, Google Scholar, and Scopus AI for Analysis & Discussion part.	"OLAP process", "OLAP use case", "OLAP advantages and disadvantages"	1. NetSuite.com. (2022, March 20). *What Is OLAP? OLAP Defined*. Oracle NetSuite. https://www.netsuite.com/portal/resource/articles/erp/online-analytical-processing-olap.shtml 2. *Exploring the Top OLAP Use Cases for Effective Data Analysis*. (2024, June 3). Exploring the Top OLAP Use Cases for Effective Data Analysis. https://risingwave.com/blog/exploring-the-top-olap-use-cases-for-effective-data-analysis/ 3. Abdulghani, A. A. (2009). Computation of OLAP data cubes. In *IGI Global eBooks* (pp. 286–292). https://doi.org/10.4018/978-1-60566-010-3.ch046	Complete OLAP Discussion part based on key points obtained from AI tools.
4/4/2026	Chua Jia Lin	Search for OLAP information using ChatGPT, Google Scholar, and Scopus AI for Recommendations part.	"OLAP recommendations"	1. Inmon, W. H. (2005). *Building the data warehouse*. John wiley & sons. 2. Chaudhuri, S., & Dayal, U. (1997). An overview of data warehousing and OLAP technology. *ACM SIGMOD Record*, *26*(1), 65–74. https://doi.org/10.1145/248603.248616	Complete OLAP Recommendations part based on key points obtained from AI tools.
6/4/2026	Cheryl Cheong Kah Voon	Search for OLTP information using ChatGPT, Gemini, Google Scholar for	"How OLTP work", "Advantage and Challenge of	1. Conn, S. S. (2005, April). OLTP and OLAP data integration: a review of feasible implementation methods and architectures for real time data analysis.	Complete OLTP Analysis& Discussion and Recommendations

		Analysis and discussion part and Recommendation part	OLTP", "Recommendation of OLTP"	In *Proceedings. IEEE SoutheastCon, 2005*. (pp. 515-520). IEEE. 2. Elmasri, R., & Navathe, S. B. (2016). Fundamentals of database systems seventh edition. 3. Nayak, S. (2025, March 13). How OLTP, OLAP, and ETL drive modern cloud data Management. CloudOptimo. https://www.cloudoptimo.com/blog/how-oltp-olap-and-etl-drive-modern-cloud-data-management/ 4. Silberschatz, A., Korth, H. F., & Sudarshan, S. (2002). Database system concepts (Vol. 5). New York: McGraw-Hill. 5. Connolly, T. M., & Begg, C. E. (2005). Database systems: a practical approach to design, implementation, and management. Pearson Education. 6. Ramakrishnan, R., & Gehrke, J. (2003). Database management systems third edition.	part based on key points obtained from AI tools.
10/4/2026	Cheryl, Chua Jia Lin	Summarise the report using ChatGPT. Then, complete the Conclusion part of the report.	"Summarise the report and give key takeaways."	N/A	Complete Conclusion part based on key takeaways

					obtained from AI tools.
10/4/2026	Chua Jia Lin	Summarise the report using ChatGPT. Then, complete the Executive Summary part of the report.	"Summarise the report and give key takeaways."	N/A	Complete Executive Summary part based on key takeaways obtained from AI tools.
11/4/2026	Cheryl, Chua Jia Lin	Formatting the report, and double check the report before submitting.	N/A	N/A	The report is well formatted.